南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name： **Intelligent Robotics**

Dept.： **Department of Computer Science and Engineering**

Exam Duration： **2 hours** **Exam Paper Setter：** **Qi Hao**

Question No.	1	2	3	4	5	6	7	8	9	10
Score	20	30	40	20						

This exam paper contains **4** questions and the score is **110** in total. (Please hand in your exam paper, answer sheet, and your scrap paper to the proctor when the exam ends.)

Problem I Multiple Choice (20 points) (only one correct answer)

1. **(2 points)** The mapping technologies DO NOT include _____.
   A. landmarks mapping
   B. reflection mapping
   C. occupancy mapping
   D. topology mapping

2. **(2 points)** Grid based maps are more preferred than landmark based maps mainly because _____.
   A. they can provide more environment information
   B. they are easy to build
   C. they do not need Gaussian assumptions
   D. they can represent moving obstacles

3. **(2 points)** The major differences between occupancy maps and reflection maps include_____.
   A. using scan data
   B. using inverse sensor models
   C. using log-odds for information accumulation
   D. using Gaussian assumptions

4. **(2 points)** Landmarks DO NOT include _____.
   A. satellites
   B. QR codes
   C. lines and faces from point clouds
   D. feature points from images

5. **(2 points)** The major approaches to improve offline SLAM performance DO NOT include _____.
   A. using high resolution sensors
   B. loop detection and closure
   C. landmark integration
   D. back-end re-optimization

6. **(2 points)** The landmark integration will be performed when _____.
   A. the likelihood of measurements fitting to any landmark is low
   B. the likelihood of the ego-robot being close to any landmark is low
   C. the total number of landmark is known
   D. the total number of landmark is unknown

7. **(2 points)** The loop closure will be performed when _____.
   A. the same landmark is detected by the ego-robot again
   B. the same grid is detected by the ego-robot again
   C. the ego-robot returns to the same location
   D. the ego-robot returns to the same orientation

8. **(2 points)** SLAM assumes that _____.
   A. robot poses are relative
   B. landmark poses are absolute
   C. measurements of landmarks are absolute
   D. odometry readings are absolute

9. **(2 points)** The steps of EKF based SLAM DO NOT include _____.
   A. association/integration
   B. prediction/estimation
   C. correction/fusion
   D. learning/modeling

10. **(2 points)** The use of fast SLAM 2.0 is more preferred than the use of fast SLAM NOT when _____.
   A. landmark based maps are used
   B. high sampling efficiency is required
   C. high resolution sensors are used
   D. high localization accuracy is demanded

Problem II Calculation (30 Points)

1. **(5 points)** Given the navigation function of $f(v_x, v_y) = 0.5v_x^2 - v_x - v_y$, and four candidate combinations between $v_x = \{1, -1\}$ and $v_y = \{1, -1\}$, what is the optimal solution of the velocity?

2. **(5 points)** Given three landmarks at the coordinates of {0, 1}, {1,0} and {1,1}, respectively, and the measurements are {1, 1, 10}, what is the localization of the ego-robot?

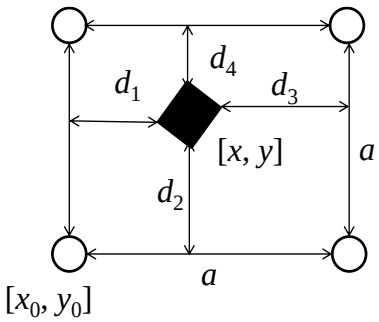
3. **(5 points)** If the weights of two robot state samples, $\{x^{[1]}, x^{[2]}\}$, are {0.9, 0.1}, and the random initialization of the selector is 0.2, then what will be the two samples after the re-sampling?

4. **(5 points)** After one round of range sensing, the number of hitting and missing, $N_{\text{hit}} = 50$ and $N_{\text{miss}} = 50$, for one cell are obtained respectively, what is the occupancy probability value for that cell with the inverse sensor model ($\rho_0 = 0.1$, $\rho_1 = 0.7$)?

5. **(5 points)** After two rounds of range sensing, the numbers of hitting and missing for one cell are $N_{\text{hit1}} = 60$, $N_{\text{hit2}} = 50$ and $N_{\text{miss1}} = 40$, $N_{\text{miss2}} = 50$, what is the reflection probability value for that cell (*assuming* $p_{\text{prior}} = 0.6$)?

6. **(5 points)** After two steps of measurements, the means and variances of a landmark location are $\mu_1 = 1$, $\Sigma_1 = 1$, $\mu_2 = 2$, $\Sigma_2 = 1$, what is the final estimation of the landmark location in terms of mean and variance?

Problem III SLAM (40 points)



1. **(5 points)** There are four landmarks in a distance of a and the coordinate of the left-lower one is $[x_0, y_0]$, the coordinate of a robot location is $\mathbf{x} = [x, y]$, and the the measurements is $\mathbf{z} = [d_1, d_2, d_3, d_4]$. What is the map model $\mathbf{m}$, where $\mathbf{m}$ is a linear vector parameterized with $[x_0, y_0, a]$? What is the measurement model with the Gaussian noise of $\mathbf{v} = [v_1, v_2, v_3, v_4]$?

2. **(5 points)** If the odometry reading is the robot displacements, $\mathbf{u} = [\Delta x, \Delta y]$, and the Gaussian motion noise is $\mathbf{w} = [w_1, w_2]$, what is the motion model for the robot?

3. **(5 points)** How to use the kalman filter to estimate landmark parameters (mean and variance of $\mathbf{m}$, that is, $\{\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m\}$), if the mean and variance of the robot location $\mathbf{x}$ ($\{\boldsymbol{\mu}_x, \boldsymbol{\Sigma}_x\}$) are given?

4. **(5 points)** How to use the kalman filter to estimate the locations of the robot ($\{\boldsymbol{\mu}_x, \boldsymbol{\Sigma}_x\}$) if the landmark parameters ($\{\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m\}$) and odometry readings ($\{\boldsymbol{\mu}_u, \boldsymbol{\Sigma}_u\}$) are given?

5. **(10 points)** How to use the fast SLAM 1.0 to estimate the locations of the robot and the landmarks simultaneously?
   (*please specifically write down the target function, the proposal function, the procedures of sampling, weighting, re-sampling*)

6. **(10 points)** How to use the fast SLAM 2.0 to estimate the robot locations robot and the grid map, respectively?
   (*please specifically write down the target function, the proposal function, the procedures of sampling, weighting, re-sampling*)

Problem IV System Design (20 points)

An autonomous robot control system has a number of components: (1) *mapper*: building a grid-map, $\mathbf{m}$, for the environment; (2) *planner solver*: formulating the motion policy, $\boldsymbol{\pi}$, for the given goal and the grid-map; (3) *initializer*: initializing the ego-robot state, $\mathbf{x}_0$; (4) *planner*: deriving the action, $\mathbf{u}_{t+1}$ for the given estimated robot state, $\mathbf{x}_t$; (5) *predictor*: using the odometry reading, $\hat{\mathbf{u}}_{t+1}$, to predict the robot state, $\mathbf{x}_{t+1}$; (6) *estimator*: updating the robot state, $\mathbf{x}_{t+1}$, with the sensor measurement, $\mathbf{z}_{t+1}$, and the map $\mathbf{m}$; then go back to (4) until the robot reaches the goal (*goal checker*).

(1) **(7.5 points)** Please draw a system diagram for (I) **robot and environment models in simulation** and (II) **controller in hardware**, where the controller has the following components: (*i*) **mapper**, (*ii*) **planner solver**, (*iii*) **initializer**, (*iv*) **planner**, (*v*) **predictor**, (*vi*) **estimator**, and (*vii*) **goal checker**.

(2) **(2.5 points)** Please draw the graph diagram of states $[\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}]$, odometry reading $[\hat{\mathbf{u}}_{t-1}, \hat{\mathbf{u}}_t, \hat{\mathbf{u}}_{t+1}]$, sensor measurements $[\mathbf{z}_{t-1}, \mathbf{z}_t, \mathbf{z}_{t+1}]$, and the grid map, $\mathbf{m}$, and the motion policy, $\boldsymbol{\pi}$.

(3) **(5 points)** Please give an example of motion policy formulation.

(4) **(5 points)** Please give an example of adaptive sampling for particle filtering.